

Planning Constructions and Justifying Them

Answer Key

High-school geometry

Quick Answers

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|--------------------------------------|---|
| 1. (a) the length of $AB = 10$, — | 7. — |
| 2. (a) $PM = 5.83$, (b) $MQ = 5.83$ | 8. (a) the distance from the centre to $A = 2.24$, |
| 3. — | (b) the distance from the centre to the image = |
| 4. — | 6.71, — |
| 5. (a) $RS = 5$, (b) $RT = 5$, — | 9. (b) the distance from A to the apex = 6, — |
| 6. — | 10. (c) the side length = 5, — |
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What is verified

9 of 19 answers machine-checked against SymPy, across 10 problems.

— marks an answer only you can judge — problems 1, 3, 4, 5, 6, 7, 8, 9, 10. The worked solution states what a correct response must contain.

Curriculum

Level: High-school geometry

HSG-CO.D — *problems 1, 2, 3, 4, 5, 6, 7, 9, 10*

HSG-SRT.A.1 — *problem 8*

Difficulty 1–5 of 5 across 19 tagged checks.

- 1.** $A(1, 2)$, $B(7, 10)$. (a) Length of $\overline{AB}$. (b) Plan for copying it onto a ray from C .

Solution (a): The horizontal run is $7 - 1 = 6$ and the vertical rise is $10 - 2 = 8$, so the length is $\sqrt{36 + 64} = \sqrt{100}$. $AB = 10$

Part (b) — instructor-judged. A full-credit plan has three moves: (1) draw a ray from C ; (2) open the compass to span exactly from A to B and do not change it; (3) put the compass point on C , strike an arc across the ray, and mark the crossing as the second endpoint. Deduct for any step that measures with a ruler — the unchanged compass span is what guarantees the copy is congruent, and a ruler reading only guarantees it to the nearest millimetre.

- 2.** Is $M(1, 4)$ the midpoint of $\overline{PQ}$ with $P(-4, 1)$ and $Q(6, 7)$?

Solution: From P to M the run is 5 and the rise is 3, so $PM = \sqrt{25 + 9} = \sqrt{34}$.

$PM = 5.83$

From M to Q the run is 5 and the rise is 3 again, so $MQ = \sqrt{34}$ as well.

$MQ = 5.83$

The two halves are equal, and M lies on $\overline{PQ}$ because the run and rise are proportional, so the claim holds.

3. Justify why $\overleftrightarrow{PQ}$ is the perpendicular bisector of $\overline{AB}$.

Instructor-judged. A model answer: both arcs were struck with the *same* compass setting, so $PA = PB$ and $QA = QB$. Each of P and Q is therefore equidistant from A and from B , and every point equidistant from two points lies on the perpendicular bisector of the segment joining them; two such points determine that line. The congruent-triangles version is equally good: $\triangle APQ \cong \triangle BPQ$ by SSS (two pairs of equal radii plus the shared side $\overline{PQ}$), so the angles at the crossing point are congruent, and being congruent and supplementary they are right. No credit for describing the drawing again without saying why the equal radii force the result.

4. Plan the bisector of $\angle ABC$, and say what each arc guarantees.

Instructor-judged. A model plan: (1) compass point at B , strike one arc crossing both sides, at D and E ; (2) compass point at D , then at E , striking two arcs of equal radius that meet inside the angle at F ; (3) draw ray $\overrightarrow{BF}$. What the arcs guarantee: the first gives $BD = BE$; the second gives $DF = EF$. With the shared side $\overline{BF}$ that is SSS, so $\triangle BDF \cong \triangle BEF$ and the two halves of the angle are congruent. Deduct for a plan that produces the ray but never says what the equal radii buy — the plan and the reason are the two halves of this question.

5. $R(5, -1)$, $S(1, 2)$, $T(9, 2)$. (a) RS . (b) RT . (c) What does it show?

Solution (a): Run 4, rise 3, so $RS = \sqrt{16 + 9}$.

$RS = 5$

Solution (b): Run 4, rise 3 again, so $RT = \sqrt{16 + 9}$.

$RT = 5$

Part (c) — instructor-judged. The two distances are equal, so R is equidistant from S and T ; every such point lies on the perpendicular bisector of $\overline{ST}$. (Here $\overline{ST}$ is horizontal with midpoint $(5, 2)$, and indeed R sits directly below it.) No credit for repeating the two distances with no statement about where R must lie.

6. Justify why the copy-an-angle construction produces a congruent angle.

Instructor-judged. A model answer: let the first arc cross the original angle's sides at D and E , and the equal-radius arc cross the new ray at D' . Then $BD = BE = B'D' = B'E'$, because one radius was used for both arcs. The span carried over unchanged gives $DE = D'E'$. So $\triangle BDE \cong \triangle B'D'E'$ by SSS, and the copied angle is congruent to the original as a corresponding part of congruent triangles. Full credit needs the three congruent pairs and the criterion; restating the steps with no congruence criterion earns nothing.

7. Plan the line through P parallel to ℓ , and name the theorem.

Instructor-judged. A model plan: (1) draw any line through P that meets ℓ ; call the crossing point X — this is the transversal; (2) copy the angle that the transversal makes with ℓ at X , placing the copy at P on the same side of the transversal and opening the same way; (3) draw the line along the copied ray. The theorem: if corresponding angles are congruent, the lines are parallel (the converse of the corresponding-angles theorem). Accept alternate interior angles if the copy is placed on the other side. Deduct for a plan that omits the theorem, and for one whose only instruction is "draw a parallel line".

8. Dilation, centre $O(0, 0)$, factor 3, $A(2, 1)$. (a) OA . (b) OA' . (c) What does the ratio show?

Solution (a): $OA = \sqrt{4 + 1} = \sqrt{5}$.

$OA = 2.24$

Solution (b): The image is $A'(6, 3)$, so $OA' = \sqrt{36 + 9} = \sqrt{45} = 3\sqrt{5}$.

$OA' = 6.71$

Part (c) — instructor-judged. The ratio $OA' : OA$ is 3, matching the scale factor. A dilation multiplies every distance from the centre by the scale factor, which is exactly why dividing the two distances recovers it. A complete answer adds that O , A and A' are collinear — a dilation moves each point along its own ray from the centre. Reduced credit for the ratio and the factor with no collinearity remark.

9. Equilateral triangle on $\overline{AB}$. (a) Plan and justify. (b) Check the apex distance for $A(0, 0)$, $B(6, 0)$, apex $(3, 5.196)$.

Part (a) — instructor-judged. A model plan: open the compass to the full span AB ; strike an arc centred at A ; without changing the span, strike an arc centred at B ; join either crossing point C to A and to B . The justification: the first arc has radius AB , so $AC = AB$; the second arc has the same radius, so $BC = AB$. Both new sides equal the original, hence equal each other, so all three sides are congruent. Deduct for the right drawing with no radius argument.

Solution (b): Run 3, rise 5.196, so the distance is $\sqrt{9 + 27} = \sqrt{36}$.

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That matches $AB = 6$, as the radius argument in part (a) predicts.

10. Square on $\overline{AB}$. (a) Plan. (b) Justify. (c) Side length for $A(2, 1)$, $B(6, 4)$.

Part (a) — instructor-judged. A model plan: (1) construct a perpendicular to $\overline{AB}$ at A , and another at B — the perpendicular-bisector construction of problem 3, applied at an endpoint; (2) set the compass to the span AB and, without changing it, mark a point at that distance up each perpendicular; (3) join the two new points with the straightedge. Any plan that reaches a square through a stated sequence of the constructions on this sheet earns full credit; a plan that names no earlier construction does not.

Part (b) — instructor-judged. Both halves are required. Equal sides: the compass span was never changed, so all four sides are congruent. Right angles: the two side lines were constructed *as* perpendiculars to $\overline{AB}$, giving three right angles, and the fourth follows because the angles of a quadrilateral sum to 360° . An answer that argues only equal sides has built a rhombus, not a square — say so on the paper and award half credit.

Solution (c): Run 4, rise 3, so $AB = \sqrt{16 + 9}$, and every side of the square must match it.

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